BCSE497J - Project-I

MEMORA: FACE-MATCHED EVENT GALLERIES

23BKT0053 SAI VARDHAN REDDY

23BCB0043 SRI RITHVIK

Under the Supervision of

Project Guide Name Dr. Kaspar S

Designation

School of Computer Science and Engineering (SCOPE)

School of Computer Science and Engineering



September 2026

ABSTRACT

Large events such as weddings, college functions, conferences, and community gatherings generate thousands of photographs and videos. Finding personal memories from such a large collection is difficult and time-consuming because guests normally have to browse through large albums or depend on folders and photographer-provided links. Memora: Face-Matched Event Galleries is an AI-powered event memory platform designed to

automate this process.

Memora uses a QR-based guest onboarding flow and multi-angle selfie capture to create a robust, normalized ArcFace centroid embedding for each guest. This allows the system to create a reliable representation of the guest's face across different poses and lighting conditions.

Uploaded event photographs and videos are processed through a FastAPI backend and Celery worker pool. The image-processing pipeline uses CLAHE preprocessing and tiled multi-scale face detection using InsightFace SCRFD. Detected faces are converted into 512-dimensional ArcFace embeddings and compared with registered guest embeddings using FAISS inner-product similarity search. A cosine-similarity threshold greater than 0.35 is used for identifying matching guests.

For video processing, Memora samples video frames at 1 FPS and identifies guest appearances. This allows the system to generate timestamped video moments while reducing unnecessary video-processing overhead.

After successful matching, Memora automatically creates personalized galleries for guests. It also provides timestamped video moments, automated email alerts, and an animated, theme-able Memory Reel. These features allow guests to easily access their personal memories without manually searching through thousands of event photographs and videos.

The system uses Next.js 14 for the frontend, FastAPI for backend services, Celery and Redis for asynchronous processing, OpenCV for preprocessing, InsightFace SCRFD and ArcFace for face detection and recognition, FAISS for vector similarity search, PostgreSQL/Supabase for database services, and Cloudinary for photo and video storage. The frontend is deployed using Vercel, while backend and worker services are deployed using Render.

Keywords: Face Recognition, ArcFace, SCRFD, InsightFace, FAISS, FastAPI, Celery, Redis, Computer Vision, Personalized Event Gallery, Event Media Processing, Memory Reel.

TABLE OF CONTENTS

Sl.No	Contents	Page No.
	Abstract	i
1.	INTRODUCTION	1
	1.1 Background	1
	1.2 Motivations	1-2
	1.3 Scope of the Project	2
2.	PROJECT DESCRIPTION AND GOALS	3
	2.1 Literature Review	3-4
	2.2 Research Gap	4
	2.3 Objectives	4-5
	2.4 Problem Statement	5
	2.5 Project Plan	6-7
3.	TECHNICAL SPECIFICATION	8
	3.1 Requirements	8-9
	3.1.1 Functional	8-9
	3.1.2 Non-Functional	9
	3.2 Feasibility Study	10-11

	3.2.1 Technical Feasibility	10
	3.2.2 Economic Feasibility	10
	3.2.2 Social Feasibility	11
	3.3 System Specification	11-12
	3.3.1 Hardware Specification	11-12
	3.3.2 Software Specification	12
4.	DESIGN APPROACH AND DETAILS	12
	4.1 System Architecture	12-14
	4.2 Design	15-16
	4.2.1 Data Flow Diagram	15
	4.2.2 Use Case Diagram	16
	4.2.3 Class Diagram	16
	4.2.4 Sequence Diagram	17
5.	REFERENCES	18

1. INTRODUCTION

1.1 Background

The rapid growth of digital photography and videography has resulted in the generation of huge amounts of media at large-scale events. Weddings, college functions, conferences, corporate events, festivals, and community gatherings may involve multiple photographers and generate thousands of photographs and videos.

After an event, guests usually need to manually browse through large collections of photographs to find images in which they appear. When photographs are collected from multiple photographers, the problem becomes even more difficult because the media is large in volume and may not be organized according to individual guests.

Traditional methods such as folder browsing, album selection, filenames, hashtags, and manual searching are inefficient for large event collections. Guests may spend considerable time searching for their own photographs.

Advances in artificial intelligence and computer vision provide an opportunity to automate this process through face detection and face recognition.

Memora: Face-Matched Event Galleries is designed as an AI-powered event memory platform that automatically detects, indexes, and matches guest faces across large-scale event photography and video.

The system provides a QR-based guest onboarding process. Guests register for an event and capture selfies from multiple angles. These images are processed to generate a normalized ArcFace centroid embedding.

Event photographs and videos uploaded by photographers are then processed using an asynchronous media-processing pipeline. CLAHE preprocessing is applied to improve image quality in challenging lighting conditions. Tiled multi-scale face detection using InsightFace SCRFD is then used to identify faces in high-resolution photographs.

The detected faces are converted into 512-dimensional ArcFace embeddings. These embeddings are compared against the registered guest embeddings using FAISS similarity search.

The main parameters of the system include 512-dimensional embeddings, three detection scales per image tile, 1 FPS video sampling, and a cosine-similarity threshold greater than 0.35.

1.2 Motivation

Large events generate thousands of photographs and videos, making it difficult for individual guests to find their personal memories.

The traditional approach requires guests to manually browse through large event albums. This becomes particularly difficult when photographs are uploaded by multiple photographers.

The primary motivation of Memora is to eliminate this manual searching process.

Instead of browsing through thousands of photographs, a guest can register their face using a QR-based onboarding process. The system then automatically searches the event media and identifies photographs and video moments containing the registered guest.

A single selfie may not provide a sufficiently robust representation of a person's face because the guest may appear in photographs with different poses, facial angles, lighting conditions, and distances from the camera.

Therefore, Memora uses multi-angle selfie capture to create a more robust face representation.

Another motivation is efficient processing of large media collections. Processing thousands of photographs and videos can require significant computational resources. Memora uses FastAPI, Celery, and Redis to create an asynchronous processing pipeline so that media can be processed in the background without blocking the main application.

The project also aims to provide a complete event-memory experience. Instead of only returning photographs, the system provides personalized galleries, timestamped video moments, automated email notifications, and an animated Memory Reel.

1.3 Scope of the Project

The scope of the Memora project includes the design, development, deployment, and testing of an AI-powered web application for automatically identifying guests in event photographs and videos.
The major features included in the project are:

1. QR-based guest registration.
2. Multi-angle selfie capture.
3. Guest face embedding generation.
4. Event creation and management.
5. Photographer media upload.
6. Large-scale photograph processing.
7. Video processing.
8. CLAHE image preprocessing.
9. Tiled multi-scale face detection.

10. InsightFace SCRFD face detection.
11. ArcFace face embedding generation.
12. 512-dimensional face embeddings.
13. FAISS vector similarity search.
14. Personalized guest galleries.
15. Timestamped video moments.
16. Automated email notifications.
17. Animated Memory Reel generation.
18. Cloud-based media storage.
19. Asynchronous background processing.

The system is designed to process media uploaded by multiple photographers and organize the resulting memories according to individual events.
The Memora project is identified as **TRL 7 – System prototype demonstration in an operational environment**. The presentation states that the application is fully developed and functional, deployed using Vercel for the frontend and Render for backend and worker services, and tested across different devices, browsers, and internet networks.

2. PROJECT DESCRIPTION AND GOALS

2.1 Literature Review

The literature review focuses on research related to face recognition, face detection, low-light image enhancement, facial occlusion, and scalable similarity search.
The reviewed research provides technical background for improving the robustness, efficiency, and scalability of the Memora face-matching pipeline.
communication system.

2.1.1 TOPOFR – TOPOLOGY ALIGNMENT ON FACE RECOGNITION

TopoFR focuses on improving the generalization and robustness of face-recognition embeddings across different facial conditions.
The methodology uses topology-alignment loss to better align the embedding space with the underlying facial structure.
The main advantage of the approach is improved embedding generalization and recognition robustness.
However, it adds complexity to the face-recognition training process.
The research provides a potential upgrade path for the ArcFace/InsightFace recognition backbone used in Memora.

2.1.2 FDLITE – A SINGLE STAGE LIGHTWEIGHT FACE DETECTOR NETWORK

FDLite focuses on developing a lightweight face detector while maintaining good detection accuracy, especially for challenging images.
The methodology uses a lightweight backbone with dual multi-task learning.
The main advantage is that the model is lightweight and suitable for efficient processing.
However, lightweight models may require careful optimization to maintain recognition and detection accuracy.
The research supports Memora's requirement for efficient face detection in high-resolution event photographs containing small faces.

2.1.3 LOW-FACENET – FACE RECOGNITION-DRIVEN LOW-LIGHT IMAGE ENHANCEMENT

Low-FaceNet focuses on improving face recognition performance in poorly lit images.
The method uses a contrastive-learning-based enhancement network to improve image brightness and recognition accuracy.
The main advantage is improved recognition in low-light conditions.
The disadvantage is that an additional image-enhancement stage increases the processing pipeline.
This research is relevant to Memora because event venues may contain poorly lit areas and changing lighting conditions.

2.1.4 MASKED FACES SURVEY – RECOGNITION, DETECTION AND UNMASKING

The Masked Faces Survey studies deep-learning approaches for face recognition under occlusion.
The research reviews methods related to recognition, detection, and unmasking of partially occluded faces.
It provides guidance for improving robustness when faces are partially covered.
Masks, sunglasses, accessories, and other forms of occlusion can reduce face-recognition performance.
This research is relevant to Memora because guests may appear in event photographs with partially occluded or obstructed faces.

2.1.5 BANG – BILLION-SCALE APPROXIMATE NEAREST NEIGHBOR SEARCH USING A SINGLE GPU

BANG focuses on large-scale approximate nearest-neighbor search.
The research is relevant to Memora because the platform needs to compare detected face embeddings against registered guest embeddings efficiently.
Large-scale similarity search becomes important when the number of registered guests and detected faces increases.

The research provides a possible direction for future optimization of vector similarity search.

The literature review topics and findings are based on the sources listed in the Memora presentation
.

2.2 Research Gap

Existing event photography systems commonly depend on manual album browsing, folder organization, or other methods of searching for photographs.
There is a need for a complete system that combines guest registration, face recognition, large-scale media processing, personalized galleries, video timestamps, notifications, and event-memory generation.
The following research gaps are identified.

2.3 Objectives

The main objectives of the Memora project are:

1. To design and develop a multi-angle face registration system that creates robust face embeddings for accurate guest identification.
2. To develop a tiled, multi-scale face detection pipeline for identifying small, angled, and poorly lit faces in high-resolution event photographs.
3. To implement an asynchronous media-processing pipeline using FastAPI, Celery, and Redis for scalable and non-blocking processing of large numbers of photographs and videos.
4. To generate 512-dimensional ArcFace face embeddings for detected faces.
5. To implement FAISS-based vector similarity search for efficient matching between detected faces and registered guests.
6. To automatically create personalized galleries for registered guests.
7. To identify guest appearances in videos and provide timestamped video moments.
8. To send automated email notifications when matching photographs and memories are available.
9. To generate animated and personalized Memory Reels.
10. To evaluate the system using face-matching, large-scale media processing, difficult-face detection, and video-processing experiments.

The project also identifies the following Sustainable Development Goals:
SDG 9 – Industry, Innovation and Infrastructure: Promoting innovation in event-media management through AI-powered face recognition, scalable processing, and intelligent search infrastructure.
SDG 11 – Sustainable Cities and Communities: Supporting efficient and accessible digital management of memories and media generated from large-scale community events.

2.4 Problem Statement

Large-scale events generate thousands of photographs and videos. These media files are often collected from multiple photographers and may contain many guests in different poses, lighting conditions, and locations.

Guests normally need to manually search through the event gallery to locate their own photographs. This process is time-consuming and becomes impractical when the number of photographs is very large.

Traditional approaches such as folders, filenames, albums, and hashtags do not provide an efficient guest-specific search mechanism.

Therefore, there is a need for an automated system that can identify registered guests across a large collection of event photographs and videos.

The proposed Memora platform addresses this problem using artificial intelligence and computer vision.

The guest first joins an event through a QR code and registers using multi-angle selfie capture. The system generates a robust face embedding for the guest.

When photographers upload event photographs and videos, the media is processed asynchronously through FastAPI, Celery, and Redis.

CLAHE preprocessing is applied to photographs. Tiled multi-scale SCRFD face detection is then used to identify faces.

Each detected face is converted into a 512-dimensional ArcFace embedding.

The detected embeddings are compared with registered guest embeddings using FAISS similarity search.

When a match is found, the corresponding photograph is added to the guest's personalized gallery.

For videos, frames are sampled at 1 FPS and matching appearances are identified with timestamps.

The system then provides personalized galleries, timestamped video moments, email notifications, and a Memory Reel.

Therefore, Memora provides an automated and personalized solution for discovering event memories.

2.5 Project Plan

Phase	Major Activities	Expected Outcome
Phase 1	Problem identification, requirements analysis, and literature review.	Defined problem, requirements, and research direction.
Phase 2	Design of guest registration, event management, media processing, and AI-processing architecture.	Finalized system architecture and data-processing workflow.
Phase 3	Development of the Next.js 14 interface, guest registration, dashboard, and photographer upload interface.	Functional web-based frontend.
Phase 4	FastAPI backend development and integration with Celery and Redis.	Functional backend and background-processing pipeline.
Phase 5	Integration of CLAHE preprocessing, tiled multi-scale SCRFD face detection, and ArcFace embedding generation.	Functional AI face-processing pipeline.
Phase 6	Integration of FAISS similarity search and personalized gallery generation.	Automated guest-to-photo matching and personalized galleries.
Phase 7	1 FPS video processing, timestamp generation, email notifications, and Memory Reel generation.	Complete event-memory experience.
Phase 8	System integration, testing, deployment, and documentation.	Functional deployed Memora system.

2.5 Project Plan

The project plan is represented using a Gantt chart showing the sequence of tasks to be completed during the development of the project.

Task No.	Task / Activity	WEEKS															
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	Literature survey	Literature survey															
2	Backend Implementation		Backend Implementation														
3	Frontend Implementation				Frontend Implementation												
4	Integration of all Module					Integration of all Module											
5	Testing of project								Testing of project								
6	Full Document preparation about project												Full Document preparation about project				
7	Submission of project														Submission of project		

Figure 2.1: Gantt Chart of the Project Plan

Fig. 1. Gantt chart

3. TECHNICAL SPECIFICATION

3.1 Requirements

The Memora system requires a combination of functional and non-functional requirements to provide reliable, scalable, and user-friendly event-memory processing.

3.1.1 FUNCTIONAL REQUIREMENTS

1. GUEST REGISTRATION AND AUTHENTICATION

The system must allow guests to join an event using a QR code.
The system must allow guests to register using multi-angle selfie capture.
The system must associate guest registration with the corresponding event.
The system must generate a face embedding for the registered guest.
The system must securely store guest registration and face-embedding information for event-specific matching.

---

2. EVENT AND MEDIA MANAGEMENT

The system must allow photographers and administrators to create and manage events.
The system must support uploading large numbers of photographs.
The system must support uploading videos.
The system must support media uploads from multiple photographers.
The system must organize uploaded media according to individual events.

3. FACE DETECTION AND RECOGNITION

The system must detect faces in uploaded photographs.
The system must use InsightFace SCRFD for face detection.
The system must support tiled and multi-scale face detection.
The system must generate 512-dimensional ArcFace face embeddings.
The system must compare detected face embeddings with registered guest embeddings.

4. ASYNCHRONOUS MEDIA PROCESSING

The system must process uploaded photographs and videos asynchronously.
The system must use Redis as the task/message broker.
The system must use Celery workers for background processing.
The system must apply CLAHE preprocessing.
The system must perform face detection and embedding extraction without blocking the main application.

5. PERSONALIZED GALLERY AND VIDEO MATCHING

The system must automatically create personalized galleries.
The system must associate matching photographs with the correct guest.
The system must process uploaded videos using frame-based processing.
The system must identify guest appearances in videos.
The system must generate timestamped video moments.
The system must use FAISS similarity search for efficient face matching.

6. NOTIFICATIONS AND MEMORY REEL

The system must automatically send email notifications when matching photographs are available.
The system must generate an animated personalized Memory Reel.
The system must allow guests to access their personalized photographs.
The system must allow guests to access timestamped video moments.
The system must allow guests to access their event Memory Reel.
These functional requirements are derived from the Memora presentation's detailed functional-requirement and module sections.

3.1.2 NON-FUNCTIONAL REQUIREMENTS

PERFORMANCE

The system should process large numbers of photographs and videos efficiently.
Asynchronous processing should ensure that large media-processing tasks do not block the main application.

SCALABILITY

The architecture should support increasing numbers of event photographs, videos, guests, and face embeddings.
Celery workers and Redis should allow processing tasks to be distributed and handled asynchronously.

RELIABILITY

The system should reliably process uploaded media and maintain the relationship between guests, events, photographs, videos, and detected faces.

USABILITY

The guest onboarding process should be simple.
QR-based registration should minimize the effort required to join an event.
Personalized galleries should reduce the need for manual searching.

ACCURACY

The system should provide robust face matching under different poses, lighting conditions, face sizes, and event environments.
Multi-angle selfie enrollment, CLAHE preprocessing, and tiled multi-scale detection are used to improve robustness.

MAINTAINABILITY

The system should be modular.
Frontend, backend, media processing, AI processing, database, storage, and vector search components should be independently maintainable.

COMPATIBILITY

The web application should work across modern browsers and different devices.

PRIVACY AND SECURITY

Guest registration data and face embeddings should be handled securely.
The actual implementation should define appropriate consent, retention, deletion, and access-control policies for biometric data.

3.2 FEASIBILITY STUDY

3.2.1 TECHNICAL FEASIBILITY

The Memora project is technically feasible because the required technologies are available and have been integrated into a functional prototype.
The frontend uses Next.js 14.
FastAPI is used to implement REST APIs and backend functionality.
Celery and Redis are used for asynchronous media processing.
OpenCV is used for CLAHE preprocessing.
InsightFace SCRFD is used for face detection.
ArcFace is used for generating face embeddings.
FAISS is used for similarity search.

PostgreSQL/Supabase is used for database services.
Cloudinary is used for photo and video storage.
Vercel is used for frontend deployment.
Render is used for backend and worker deployment.
The project presentation identifies the system as TRL 7 and states that it has been tested across different devices, browsers, and internet networks.

3.2.2 ECONOMIC FEASIBILITY

The Memora project is economically feasible because the architecture is based on widely available software frameworks and cloud services.
The system uses technologies such as Next.js, FastAPI, Celery, Redis, OpenCV, InsightFace, FAISS, PostgreSQL, Supabase, Cloudinary, Vercel, and Render.
The exact operational cost depends on the service tiers selected and the amount of event media processed.
The major possible cost areas include:

- Cloud media storage.
- Database services.
- Backend hosting.
- Worker processing resources.
- Media processing.
- Email delivery.
- Network bandwidth.

For an academic prototype, the selected architecture provides a practical approach for development, testing, and deployment.

3.2.3 SOCIAL FEASIBILITY

Memora is socially feasible because it addresses a practical problem faced by guests after large events.
Guests commonly have difficulty finding their photographs when an event contains thousands of photographs.
Memora allows guests to register their faces using QR-based onboarding and multi-angle selfie capture.
The system automatically finds photographs containing the guest.
Personalized galleries allow guests to access their memories without manually browsing the complete event collection.
Timestamped video moments allow guests to find their appearances in videos.
Email notifications provide an automated way of informing guests when their memories are ready.
The Memory Reel provides a more engaging method of viewing personal event memories.
Therefore, Memora can improve the accessibility and convenience of event photography and video memories.

3.3 SYSTEM SPECIFICATION

3.3.1 HARDWARE SPECIFICATION

The Memora presentation does not provide fixed hardware specifications such as processor model, RAM, GPU, or storage capacity.

Therefore, exact hardware values should be entered according to the actual systems used by the project team.

The general hardware requirements are:

- Computer or laptop for software development.
- Camera for guest selfie registration.
- Internet connectivity.
- Smartphone, tablet, laptop, or desktop for accessing the web application.
- Sufficient computational resources for development and testing.
- Cloud computing resources for backend and worker processing.

3.3.2 SOFTWARE SPECIFICATION

FRONTEND

Next.js 14

Next.js 14 is used for developing the web-based user interface.

The frontend provides the guest dashboard, administrator dashboard, photographer upload interface, QR-based guest registration, and multi-angle selfie capture.

BACKEND

FastAPI

FastAPI is used to provide REST API services.

It manages guest registration, authentication, event management, media upload, and communication with background processing services.

ASYNCHRONOUS PROCESSING

Celery

Celery is used to process photographs and videos asynchronously.

It allows computationally intensive media-processing operations to run in the background.

MESSAGE BROKER

Redis

Redis is used as the task/message broker.

It coordinates processing tasks between the FastAPI backend and Celery workers.

IMAGE PREPROCESSING

OpenCV

OpenCV is used for CLAHE preprocessing.

CLAHE improves image contrast and assists the processing of photographs under challenging lighting conditions.

FACE DETECTION

InsightFace SCRFD

SCRFD is used for face detection.

Memora uses tiled and multi-scale processing to improve detection of small and difficult faces.

FACE RECOGNITION

InsightFace buffalo_l / ArcFace

ArcFace is used to generate 512-dimensional face embeddings.

These embeddings are used to represent faces numerically for similarity matching.

VECTOR SEARCH

FAISS

FAISS is used for vector similarity search.

It compares detected face embeddings with registered guest embeddings.

DATABASE

PostgreSQL / Supabase

PostgreSQL/Supabase is used to store user, event, guest, and face metadata.

MEDIA STORAGE

Cloudinary

Cloudinary is used for storing event photographs and videos.

FRONTEND DEPLOYMENT

Vercel

Vercel is used for deploying the frontend application.

BACKEND AND WORKER DEPLOYMENT

Render

Render is used for deploying backend and worker services.

The complete technology stack is specified in the Memora architecture section of the presentation.

4. DESIGN APPROACH AND DETAILS

4.1 SYSTEM ARCHITECTURE

The Memora architecture consists of multiple layers responsible for user interaction, application services, artificial intelligence processing, data management, media storage, and deployment.

The major architectural layers are:

1. Client Layer.
2. Backend Layer.
3. AI/Processing Layer.
4. Database and Storage Layer.
5. External Services.

CLIENT LAYER

The Client Layer provides the user-facing interface of Memora.

The major components include:

• Next.js 14 User Interface.
• Guest Dashboard.
• Administrator Dashboard.
• Photographer Upload Interface.
• QR-based Guest Registration.
• Multi-angle Selfie Capture.

The guest uses the interface to join an event and register their face.
The photographer uses the upload interface to submit photographs and videos.
The administrator manages events and related media.

BACKEND LAYER

The Backend Layer provides the application services.
The major components include:

- FastAPI REST API.
- Authentication.
- Event Management.
- Guest Management.
- Media Upload APIs.
- Celery Worker Pool.
- Redis Message Broker.

FastAPI handles requests from the frontend.
Celery performs background processing.
Redis manages processing tasks and coordinates communication with Celery workers.

AI / PROCESSING LAYER

The AI/Processing Layer forms the core of the Memora system.
The major components are:

CLAHE PREPROCESSING

CLAHE preprocessing is applied to improve image contrast.
This is useful for photographs where lighting conditions make face detection difficult.

TILED MULTI-SCALE FACE DETECTION

High-resolution event photographs may contain small faces.
The system uses tiled and multi-scale SCRFD detection to identify faces at different sizes and positions.

ARCFACE EMBEDDING GENERATION

Detected faces are converted into 512-dimensional ArcFace embeddings.
These embeddings represent the facial characteristics of the detected person.

FAISS SIMILARITY SEARCH

FAISS is used to compare detected face embeddings with registered guest embeddings.
The system uses a cosine-similarity threshold greater than 0.35 for matching.

VIDEO FRAME PROCESSING

Uploaded videos are sampled at 1 FPS.
The sampled frames are processed for face detection and recognition.
Matching frames are used to generate timestamped guest appearances.

DATABASE AND STORAGE LAYER

The Database and Storage Layer consists of PostgreSQL/Supabase, Cloudinary, FAISS, and Redis.

POSTGRESQL / SUPABASE

PostgreSQL/Supabase stores:

- User information.
- Event information.

• Guest information.
• Face metadata.
• Application metadata.

CLOUDINARY

Cloudinary stores:
• Event photographs.
• Event videos.
• Other required media.

FAISS

FAISS provides vector similarity search for face embeddings.

REDIS

Redis provides task queue and processing coordination.

EXTERNAL SERVICES

VERCEL

Vercel is used for frontend deployment.

RENDER

Render is used for backend and worker deployment.

CLOUDINARY

Cloudinary provides media storage.

SUPABASE / POSTGRESQL

Supabase/PostgreSQL provides database services.

The presentation specifies these architecture components and external services.

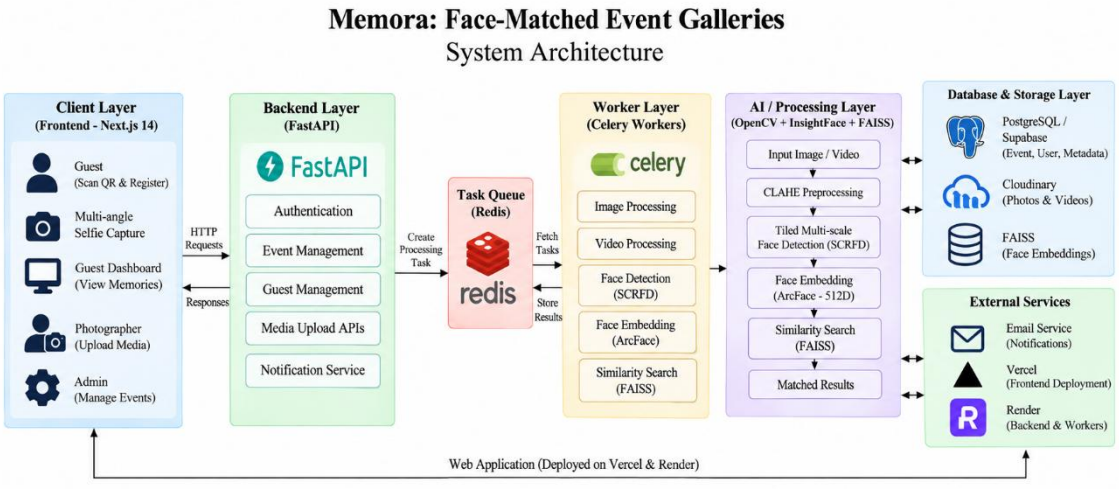


Figure 1: System Architecture of Memora

Fig. 2. System Architecture

4.2.1 Data Flow Diagram

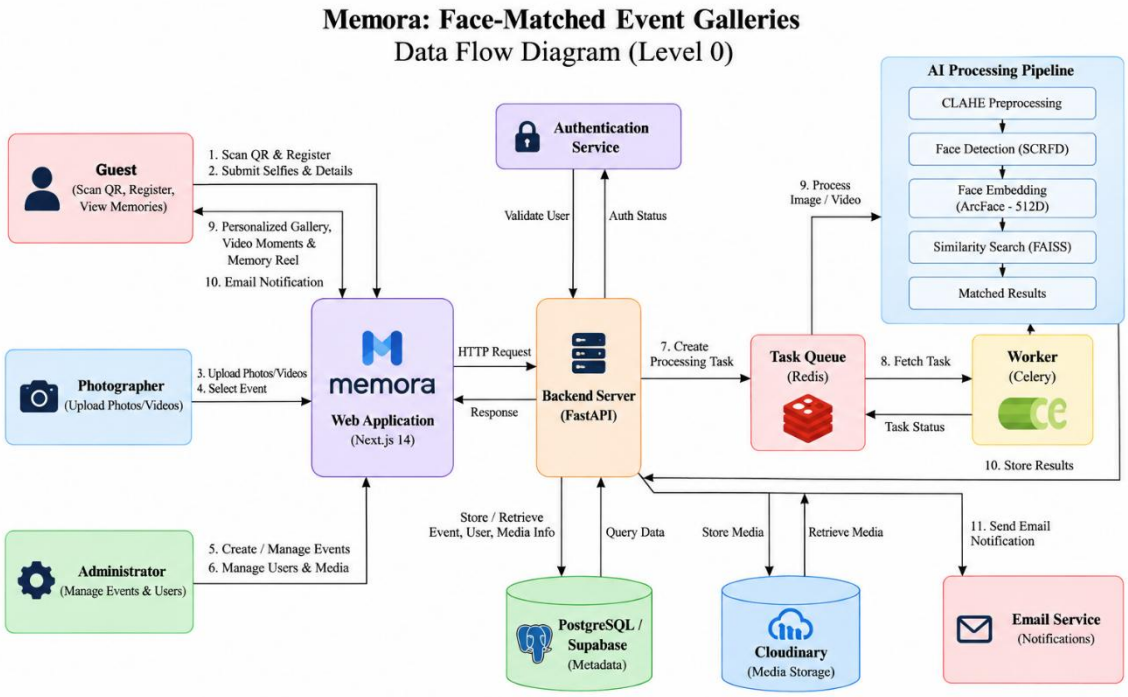


Figure 2: Data Flow Diagram of Memora

4.2.2 Use Case Diagram

Memora: Face-Matched Event Galleries

Use Case Diagram

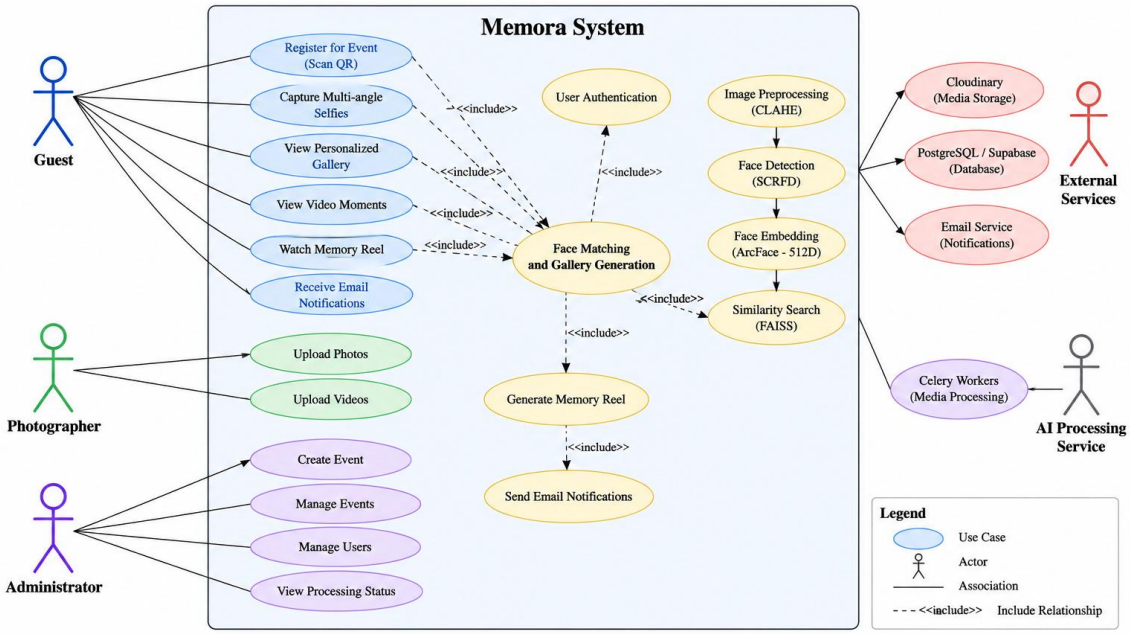


Figure 3: Use Case Diagram of Memora

4.2.3 Class Diagram

Memora: Face-Matched Event Galleries

Class Diagram

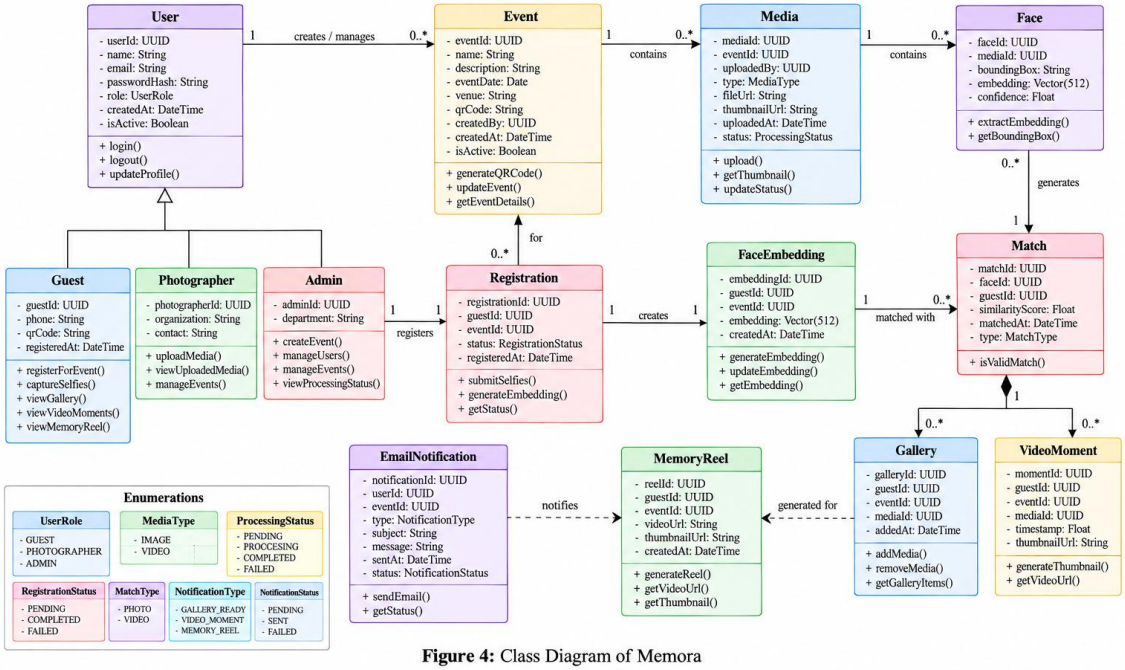


Figure 4: Class Diagram of Memora

4.2.4 Sequence Diagram

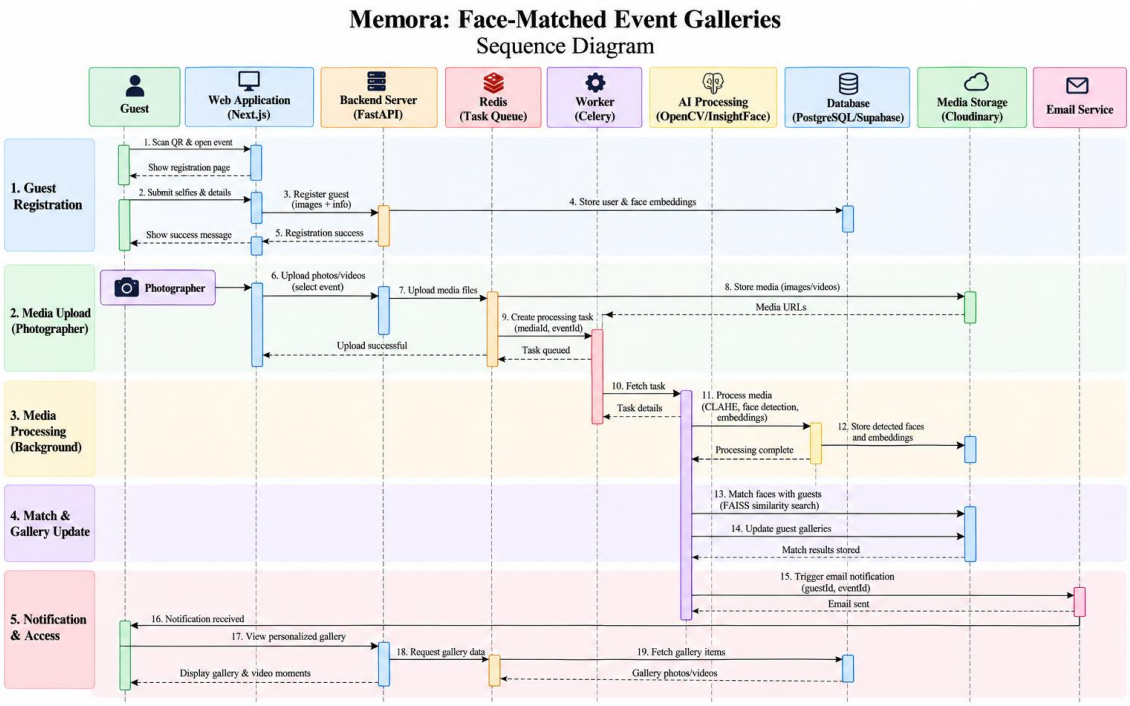


Figure 5: Sequence Diagram of Memora

5. REFERENCES

Weblinks

1. InsightFace, “InsightFace: 2D and 3D Face Analysis Project,” [Online]. Available: InsightFace Official Documentation
2. Facebook Research, “Faiss: A Library for Efficient Similarity Search and Clustering of Dense Vectors,” [Online]. Available: Faiss Documentation and Repository
3. Celery, “Celery Distributed Task Queue,” [Online]. Available: Celery Documentation
4. Next.js, “Next.js Documentation,” [Online]. Available: Next.js Documentation
5. FastAPI, “FastAPI Documentation,” [Online]. Available: FastAPI Documentation
6. OpenCV, “Open Source Computer Vision Library,” [Online]. Available: OpenCV Documentation

---

Journals

1. J. Johnson, M. Douze, and H. Jégou, “Billion-scale similarity search with GPUs,” *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2021.
2. M. Douze, A. Guzhva, C. Deng, J. Johnson, G. Szilvasy, P. Mazaré, M. Lomeli, L. Hosseini, and H. Jégou, “The Faiss library,” arXiv preprint arXiv:2401.08281, 2024.
3. S. M. Pizer et al., “Adaptive histogram equalization and its variations,” *Computer Vision, Graphics, and Image Processing*, vol. 39, no. 3, pp. 355–368, 1987.
4. J. A. Stark, “Adaptive image contrast enhancement using generalizations of histogram equalization,” *IEEE Transactions on Image Processing*, vol. 9, no. 5, pp. 889–896, 2000.
5. A. M. Reza, “Realization of the contrast limited adaptive histogram equalization (CLAHE) for real-time image enhancement,” *Journal of VLSI Signal Processing Systems*, vol. 38, pp. 35–44, 2004.

The CLAHE references are relevant because your Memora pipeline uses CLAHE as an image-preprocessing stage.

Conference

1. J. Deng, J. Guo, N. Xue, and S. Zafeiriou, “ArcFace: Additive Angular Margin Loss for Deep Face Recognition,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 4690–4699.
2. J. Guo, J. Deng, A. Lattas, and S. Zafeiriou, “Sample and Computation Redistribution for Efficient Face Detection,” in *International Conference on Learning Representations (ICLR)*, 2022.

3. F. Schroff, D. Kalenichenko, and J. Philbin, “FaceNet: A Unified Embedding for Face Recognition and Clustering,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 815–823.
4. J. Deng, J. Guo, E. Ververas, I. Kotsia, and S. Zafeiriou, “RetinaFace: Single-Shot Multi-Level Face Localisation in the Wild,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 5203–5212.
5. J. Deng, J. Guo, N. Xue, and S. Zafeiriou, “ArcFace: Additive Angular Margin Loss for Deep Face Recognition,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 4690–4699.
6. J. Guo, J. Deng, A. Lattas, and S. Zafeiriou, “Sample and Computation Redistribution for Efficient Face Detection,” *International Conference on Learning Representations*, 2022.
7. J. Deng et al., “RetinaFace: Single-Shot Multi-Level Face Localisation in the Wild,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020.

Standards / Technical Documentation

1. FastAPI, “FastAPI Technical Documentation,” [Online]. Available: FastAPI Documentation
2. Celery Project, “Celery: Distributed Task Queue,” [Online]. Available: Celery Documentation
3. OpenCV, “OpenCV-Python Tutorials and Documentation,” [Online]. Available: OpenCV Documentation
4. PostgreSQL Global Development Group, “PostgreSQL Documentation,” [Online]. Available: PostgreSQL Documentation
5. Redis, “Redis Documentation,” [Online]. Available: Redis Documentation
6. Cloudinary, “Cloudinary Documentation,” [Online]. Available: Cloudinary Documentation
7. Vercel, “Vercel Documentation,” [Online]. Available: Vercel Documentation
8. Render, “Render Documentation,” [Online]. Available: Render Documentation

---

Books

1. R. Szeliski, *Computer Vision: Algorithms and Applications*, 2nd ed., Springer, 2022.
2. I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.
3. D. A. Forsyth and J. Ponce, *Computer Vision: A Modern Approach*, 2nd ed., Pearson, 2012.
4. R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed., Pearson, 2018.

5. A. K. Jain, *Fundamentals of Digital Image Processing*, Prentice Hall, 1989.

Best version for your report

If your professor expects a reference section **similar in length to the screenshot**, I recommend using **around 20–25 references**, not all 30+ above.

A clean final arrangement would be:

Weblinks — 6 references
Journals — 5 references
Conference — 7 references
Standards / Technical Documentation — 8 references
Books — 5 references

This gives your Memora report a professional reference section similar to the sample you showed.

Also, **do not use the older eight references from the PPT blindly** if they don't actually support your implementation. For Memora, the strongest technical references are **ArcFace, SCRFD, FaceNet, RetinaFace, FAISS, and CLAHE**, because they directly relate to the face embedding, face detection, similarity-search, and preprocessing components of your system.